

SMART TRAFFIC MONITORING SYSTEM USING RFID-BASED SPEED DETECTION

ABSTRACT:

The **Smart Traffic Monitoring System Using RFID-Based Speed Detection** is an intelligent traffic management solution designed to improve road safety and reduce manual traffic monitoring. The system uses RFID technology to identify vehicles and measure their speed by recording the time taken to travel between two RFID readers placed at a fixed distance. If a vehicle exceeds the speed limit, the system automatically generates an e-challan and stores the vehicle details. A smart barrier remains closed until the fine is paid. Additionally, IR sensors monitor traffic density and dynamically adjust traffic signal timings to minimize congestion. This project provides an efficient, reliable, and cost-effective solution suitable for modern smart city applications.

AIM:

To design and implement a Smart Traffic Monitoring System using RFID-based speed detection for automated vehicle speed monitoring, e-challan generation, smart barrier control, and traffic signal management based on vehicle density.

OBJECTIVES:

- Detect vehicles using RFID technology.
 - Measure vehicle speed accurately.
 - Identify over-speeding vehicles.
 - Generate e-challans automatically.
 - Control barrier access based on payment status.
 - Reduce traffic congestion using density-based traffic control.
 - Improve road safety with minimal human intervention.
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COMPONENTS USED:

- ESP32 Microcontroller
- RFID Readers (3)
- RFID Tags
- IR Sensors

- Servo Motor (SG90)
 - LEDs (Red, Yellow, Green)
 - 220Ω/330Ω Resistors
 - NPN Transistor
 - 1N4007 Diode
 - 9V Power Supply
 - Connecting Wires and Breadboard
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WORKING PRINCIPLE:

The system works as follows:

1. Every vehicle is equipped with a unique RFID tag.
 2. The first RFID reader records the entry time of the vehicle.
 3. The second RFID reader records the exit time.
 4. The ESP32 calculates the vehicle speed using the known distance between the RFID readers.
 5. If the speed exceeds the predefined limit, the system automatically generates an e-challan.
 6. The vehicle information and violation details are stored.
 7. At the checkpoint, another RFID reader checks whether the challan has been paid.
 8. If the fine is unpaid, the servo-controlled barrier remains closed.
 9. Once payment is completed, the barrier opens automatically.
 10. IR sensors detect traffic density and adjust the green signal timing according to the number of vehicles on each road.
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APPLICATIONS:

- Smart Traffic Management
 - Automatic Speed Monitoring
 - E-Challan System
 - Toll Plaza Automation
 - Smart Parking Systems
 - Smart City Transportation
 - Highway Speed Monitoring
 - Intelligent Traffic Signal Control
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ADVANTAGES:

- Accurate vehicle identification.
- Automatic speed monitoring.
- Reduces manual traffic policing.
- Improves road safety.
- Faster fine collection.

- Intelligent traffic signal control.
 - Reduces traffic congestion.
 - Easy integration with smart city infrastructure.
 - Low maintenance and reliable operation.
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FUTURE SCOPE:

- Integration with AI-based traffic analysis.
 - Cloud-based vehicle database.
 - Mobile application for e-challan payment.
 - GPS-based vehicle tracking.
 - Automatic emergency vehicle priority.
 - Face and number plate recognition integration.
 - Real-time traffic monitoring dashboard.
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CONCLUSION:

The **Smart Traffic Monitoring System Using RFID-Based Speed Detection** successfully demonstrates an intelligent and automated solution for modern traffic management. By combining RFID technology, ESP32, IR sensors, and automated barrier control, the system accurately detects vehicle speed, identifies traffic violations, and generates e-challans without human intervention. The density-based traffic signal control further improves traffic flow and reduces congestion. This project is cost-effective, reliable, scalable, and can be implemented in smart cities to enhance road safety and efficient traffic management. It is based on the content and structure of your uploaded report with the updated title and revised matter.